

One Dial

Learning cost-aware model assignment for multi-role agent harnesses
from graded outcomes

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Abstract

A multi-role agent harness must decide, before each task, which language model occupies which role: which one writes, which one reviews, which one plans. We give a decision-theoretic account of that assignment problem and a learner for it whose only user-facing parameter is a scalar $L \geq 0$, the dollar cost of a failed task. Every other quantity — prices, token volumes, acceptance probabilities, the weight of a published catalog, the reliability of each grading source — is measured or learned. The assignment minimises $J(a; x, L) = \mathbb{E}[B(a, x)] + L(1 - P(g=1 \mid a, x))$, where g is an executable, model-free grade on the artefact the run leaves behind. We characterise exactly what the dial reaches: sweeping L over $[0, \infty)$ traces the exposed vertices of the lower convex hull of the achievable (bill, failure) set and nothing else, so Pareto-optimal assignments strictly inside the hull are selected by no stake. We prove that the scale-free knee used by a standard baseline is always such a vertex, and retire it anyway, because its implied stake is covariant with the price scale while a user’s stake is not. Around that core we give a per-role additive-logit quality model with semi-bandit feedback, a hierarchical catalog-anchored prior with a cap derived from a between-site random effect, sample complexity in closed form, drift by variance inflation, a cascade break-even condition, and a switching rule with an anytime-valid test. We evaluate on 202 graded runs of one production harness, 132 of them carrying a per-role token split. Three results. The measured role shape is nothing like the fixed one: the writing role takes a median 0.97 of a run’s tokens but only 0.52 of its dollars, while the reviewing role takes a median 0.026 of the tokens and 0.44 of the dollars, with a tail reaching 0.95 and nine runs in 132 on which the role that only reads the work takes more than half the tokens; learning that shape — shrunk on the log scale, which matters — cuts leave-one-out bill prediction error from 302% to 58% and the median error factor from $3.4\times$ to $1.6\times$. In a hindsight check over the twelve assignments run at least three times, the learned rule’s pick at $L=\$1$ costs 3.4% more than the best assignment in hindsight, against 57.5% for the same rule priced with a fixed shape; at $L=\$10$ it costs 92% more, and we show why — the additive model cannot represent the role-pair interaction that the hindsight winner exploits. Replaying the ledger in time order, the rule’s empirical regret at $L=\$1$ flattens from \$0.90 to \$0.027 per task across thirds while a fixed-shape variant stays at \$0.45; at $L=\$10$ it does not flatten at all, which is what a stake of ten dollars and 136 graded runs are jointly unable to resolve. We report the negative results in full.

Keywords: model selection, decision theory, combinatorial semi-bandits, Thompson sampling, cascades, empirical Bayes, cost-aware learning, LLM agents.

1 Introduction

An agent harness that drives language models through software tasks makes a structured choice before every task: which model occupies which role. Contemporary harnesses run two to five roles — one that writes the change, one that reviews it, one that decomposes the brief, sometimes one that grades and one that converses — and the published model catalog offers hundreds of candidates for each. The choice matters twice over. It sets the bill, which varies by two orders of magnitude across feasible assignments, and it sets the probability that the task comes back done, which varies by tens of percentage points. The two move together, and the rate at which a practitioner is willing to trade one for the other is not a property of the system.

The information available to make the choice arrives in three units and on three timescales. A published catalog gives a price and one or more benchmark indexes for every model on the day it appears — in the snapshot we use, 522 models, 417 of them priced. A site’s own past runs are few, but they are drawn from exactly the distribution that matters. Other sites’ runs are many, but they are drawn from other people’s repositories. A usable rule must combine all three, and must keep working on the day a model nobody has run appears in the feed.

The thesis. We argue for a one-parameter decision rule. The practitioner states one number: L , the dollars a failed task costs them. Everything else — the per-role token volumes, the prices, the acceptance probabilities, the weight the catalog deserves, the reliability of each source of grades — is measured from graded runs or read from a public feed. The rule minimises expected loss

$$J(a; x, L) = \mathbb{E}[B(a, x)] + L(1 - P(g=1 \mid a, x)),$$

its reachable set is exactly the exposed lower convex hull of the achievable (bill, failure) set, and as graded runs accumulate its picks converge to the assignment that minimises realised loss. The units are dollars throughout. There is no utility scale, no temperature, and no exchange rate the practitioner does not set.

Why one dial and not zero. A rule with no dial must choose the trade-off on the practitioner’s behalf, and the usual device for that is a scale-free knee: the point on the cost–quality front farthest above the chord in (log cost, quality). We show in Section 5 that such a knee is always a hull vertex, hence always the answer for *some* stake — and that this is exactly the problem, because the stake it implies is never shown, and it moves when prices move in a direction opposite to the one a fixed stake implies. A rule with no units cannot be told what a failure costs.

Why one dial and not several. Every additional knob — an exploration rate, a quality floor, a temperature, a price ceiling — is a quantity the practitioner must set without a unit in which to reason about it, and each interacts with the others. We show that the quantities usually exposed as knobs are either derivable from L and the measured cells (the exploration budget prices itself inside J ; the switching threshold is a cache loss with a receipt) or estimable from data (the catalog’s weight, the between-site variance, the drift rate, each a moment estimator).

Contributions.

1. A decision-theoretic formulation of role assignment with a single scalar in dollars, and a complete characterisation of what sweeping it can and cannot reach (Propositions 2–3), together with a proof that the scale-free knee is always reachable (Proposition 4) and the scale argument for retiring it anyway (Proposition 5).
2. An executable, model-free reward, and a derivation of what a noisy proxy for it is worth in units of graded runs (Proposition 1), measured on three candidate proxies (Section 4).
3. A per-role additive-logit quality model with semi-bandit feedback, a hierarchical catalog-anchored prior whose external evidence is capped by a between-site random effect (Proposition 6), closed-form sample complexity (Proposition 7), and drift by variance inflation (Section 7).
4. A learned per-(role, model) token shape in place of a fixed one, and the observation — which turns out to carry the entire effect — that the shape must be shrunk on the log scale (Section 8).
5. Cascades, exploration under a hard budget, switching under an anytime-valid test, and pooling across sites, each as a separate mechanism with its own break-even condition and its own stated limits (Sections 9–12).
6. An evaluation on 202 graded runs of a production harness — a cost-model experiment, a hindsight check against the best assignment at three stakes, and a convergence replay — reported with its negative results (Section 13).

2 Setting and notation

Definition 1 (Roles, models, assignments). Let $\mathcal{R}$ be a finite set of *roles* and $\mathcal{M}$ a finite set of *models*. An *assignment* is a map $a : \mathcal{R} \rightarrow \mathcal{M}$ subject to feasibility: each role r carries a predicate $\text{fit}_r(m)$ (tool use, modalities, context window) and a set $D \subseteq \mathcal{R} \times \mathcal{R}$ of role pairs that may not be held by models of the same vendor. Write $\mathcal{A}$ for the feasible set.

Throughout, $\mathcal{R} = \{\text{writer, reviewer, planner}\}$: the writer produces the change, the reviewer reads it without producing one, the planner decomposes the brief into steps. Nothing below depends on $|\mathcal{R}| = 3$, and nothing depends on those three being the roles; a harness with a separate grading role or a separate conversational role adds coordinates to a without changing any statement.

A *task* carries a context x recorded before the run starts: an entry channel $c \in \mathcal{C}$ (which interface the work arrived through), a size class $s \in \mathcal{S}$, and any other pre-run covariate. A *run* is one attempt by one assignment on one task, and leaves behind an *artefact* — in the software setting, a working tree.

symbol	meaning
$r \in \mathcal{R}, m \in \mathcal{M}$	a role; a model
$a \in \mathcal{A}, a(r)$	an assignment; the model it puts in role r
$x = (c, s, \dots)$	the task’s pre-run context: entry channel $c \in \mathcal{C}$, size class $s \in \mathcal{S}$
$g \in \{0, 1\}$	the grade: 1 iff the artefact is acceptable by the harness’s own checks
z	an auxiliary observation from a source in $\mathcal{G}$ (an independent rating, review findings, a run’s own claim)
$B(a, x)$	the run’s bill in dollars
$L \geq 0$	the <i>stake</i> : what one failed task costs this practitioner, in dollars
$J(a; x, L)$	the objective, Eq. (5)
$\theta_{r,m}$	role r ’s contribution to the log-odds of acceptance when held by m
α_x	the task’s own difficulty on the log-odds scale
φ_m	the catalog’s published indexes for m
w_r	role r ’s weighting of those indexes
κ	the catalog prior’s strength, in graded runs
$\sigma_0^2, \sigma_u^2, \tau^2$	prior variance; between-site variance; drift variance per day
$T_{r,m}$	the token shape of (r, m) : input, cached and output tokens per call, calls per task
π_m, π_m^c, χ_m	prompt, cache-read and completion price per token
$b, \beta(b)$	a log-price band; the set of models in it
ρ_{src}	source $\text{src} \in \mathcal{G}$ ’s reliability: what one of its observations is worth in grades
E	the per-site, per-day cap on exploration spend, in dollars
δ	the switching threshold: cache loss plus hysteresis

Table 1: Notation. Dollars are dollars throughout; there is no utility unit and no exchange rate the practitioner does not set.

3 Baselines

We state two baselines precisely, because the contributions below are differences from them and because both are what deployed systems actually do.

Baseline B0: a fixed token shape and a scale-free knee. Quality is read from the catalog alone. A model’s score for a role is a weighted, pool-normalised blend of its published indexes,

$$\hat{q}_r(m) = 100 \sum_i w_{r,i} \frac{\varphi_{m,i}}{\max_{m' \in \mathcal{M}} \varphi_{m',i}}, \quad (1)$$

the maximum taken over the candidate list handed to the call, so the same model scores differently in a stronger pool; a missing index is imputed from ratios to the indexes the model does publish. Cost is read from a *fixed shape*: each role carries an input-to-output ratio, a cache-hit share, and a share of the task’s tokens, all of them constants. The role’s unit cost is $(1 - \gamma_r)\pi_m + \gamma_r\pi_m^c + \chi_m/\text{InOut}_r$ and the assignment’s bill is the volume-weighted sum. Assignments are sorted by bill and swept for strict improvement in quality, giving the Pareto front $\mathcal{F}$; the chosen assignment is the member farthest above the chord of the front in $(\ln B, Q)$:

$$\text{knee} = \arg \max_{a \in \mathcal{F}} \left[(Q(a) - Q_0) - (Q_1 - Q_0) \frac{\ln B(a) - \ln B_0}{\ln B_1 - \ln B_0} \right]. \quad (2)$$

Baseline B1: a posterior-mean rule with a catalog prior. B1 keeps B0’s fixed cost shape but replaces the catalog score with a shrinkage blend of it and the site’s own measurements. A cell supplies a mean $\bar{y}_{r,m}$ over N observations, and the role’s score becomes

$$(1 - w) \hat{q}_r(m) + w \bar{y}_{r,m}, \quad w = \frac{N}{N + \kappa}, \quad \kappa = 30. \quad (3)$$

This is the posterior mean of a conjugate normal model in which the catalog is a prior worth κ observations; Section 7 makes that reading precise and Proposition 6 generalises it to three sources.

Four things neither baseline can see.

G1 No exploration. The pick is the argmin at the posterior mean. A role–model pair nobody has rated keeps its catalog score forever, and a model outside the catalog’s index coverage keeps whatever default it was given. Section 9 shows this is not merely a theoretical gap: it is the one regime in which the rule’s regret does not fall at all.

- G2** *No measured cost.* The token shape is a constant, so the bill of an assignment is a statement about prices and not about what the models do. In particular nothing in the shape varies by model: a reviewer that emits ten times as many tokens as another is priced as though it emitted the same number. Section 13 measures what that omission costs.
- G3** *Two scales under one name.* Systems of this kind routinely publish a continuous rubric score and a binary defect indicator under a single metric, and average them into one cell. A Gaussian mean over a mixture of a 0–100 rubric and a two-point Bernoulli is not a quantity. Section 4 separates them.
- G4** *Leakage in the covariates.* Where a size class is computed from the artefact’s token count *after* the run, it cannot be conditioned on before the pick. A feature known only after the outcome is not a feature.

A rule with no units, moreover, cannot be told what a failure costs, which is the motivation for everything in Section 5.

4 The reward: an executable grade, and what a proxy is worth

Definition 2 (The grade). For an artefact ℓ , the harness runs a fixed, executable battery — in the software setting: a build over the whole module; a static check; the test suites of exactly the packages the change touched, with the cache defeated; and a formatting check. Define $g(\ell) = 1$ if all succeed and 0 otherwise. The grade is a function of the artefact and the repository alone. No language model is consulted, so no model grades itself, and no model’s opinion of an artefact can move it.

This is a deliberately narrow reward. It says nothing about whether the change is the change that was asked for; Section 16 treats that as the principal threat to validity. What it buys is a reward that is cheap, reproducible by a stranger, and immune to the single largest hazard in an agent harness — that the thing being optimised is a claim made by the thing being optimised.

4.1 Auxiliary observations and their reliability

Other signals exist and are not thrown away: an independent rating in $[0, 100]$; the reviewing role’s counts of major and minor findings; the run’s own account of how it stopped. They enter as observations of g with a fitted reliability, never as g .

Proposition 1 (What one noisy binary observation is worth). *Let $g \sim \text{Bernoulli}(p)$ and let $z \in \{0, 1\}$ satisfy $P(z=1 \mid g=1) = a$ and $P(z=0 \mid g=0) = b$ with $\gamma := a + b - 1 > 0$, and $z \perp\!\!\!\perp$ everything else given g . Then $z \sim \text{Bernoulli}(q)$ with $q = ap + (1 - b)(1 - p)$, and the Fisher information z carries about p is $\gamma^2/(q(1 - q))$. Relative to one direct observation of g , which carries $1/(p(1 - p))$, the effective count of one z is*

$$\rho = \gamma^2 \frac{p(1 - p)}{q(1 - q)} \in [0, 1], \quad (4)$$

with $\rho = 1$ iff $a = b = 1$.

Proof. $q = ap + (1 - b)(1 - p)$ gives $dq/dp = a - (1 - b) = \gamma$. The Fisher information of a $\text{Bernoulli}(q)$ observation about its own parameter is $1/(q(1 - q))$; by the chain rule for Fisher information under a smooth reparametrisation, its information about p is $(dq/dp)^2/(q(1 - q)) = \gamma^2/(q(1 - q))$. Dividing by $1/(p(1 - p))$ gives (4). For the range, write $\gamma = a + b - 1$ and note $q(1 - q) \geq \gamma^2 p(1 - p)$ because q is an affine image of p with slope γ into $[0, 1]$: $\text{Var}(z) \geq \text{Var}(\mathbb{E}[z \mid g]) = \gamma^2 p(1 - p)$, the inequality being the law of total variance with the conditional variance dropped, and it is an equality exactly when $\text{Var}(z \mid g) = 0$ almost surely, i.e. $a = b = 1$. $\square$

An observation from source $\text{src} \in \mathcal{G}$ is therefore admitted with weight ρ_{src} in the count N of Eq. (3) and its successors, and $(a_{\text{src}}, b_{\text{src}})$ are fitted wherever grades and proxies co-occur — a Dawid–Skene fit over sources [4], the binary case of the many-facet Rasch calibration [17] that a rater-severity fit performs on the continuous scale.

4.2 Three proxies, measured

Figure 1 applies Eq. (4) to a ledger of 202 graded runs (Section 13 describes it), using “the run carried no major finding” as the grade, for want of an executable grader in the historical data. Three results, and the third is a design consequence.

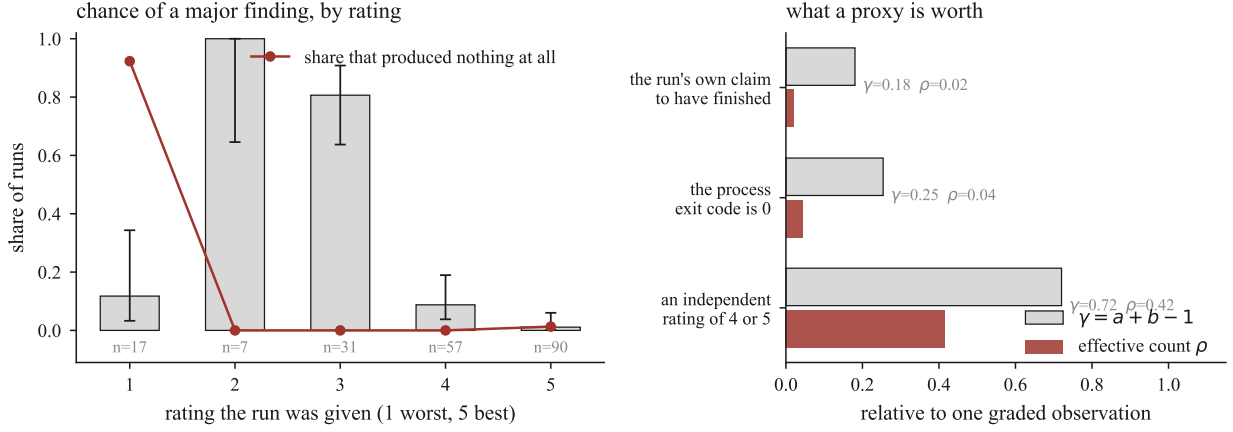


Figure 1: Measured. Left: the chance a run carried at least one major finding, by the rating it was given, with Wilson intervals; the accent line is the share of runs at that rating that produced nothing at all. The relation is not monotone, and the non-monotonicity is exactly the runs that did no work. Right: Proposition 1 applied to three candidate proxies for the graded bit. A run’s own account of itself is worth $\rho = 0.02$ of a grade.

A run’s own claim is worth almost nothing. The harness’s own verdict that a run finished has sensitivity 0.56, specificity 0.62, $\gamma = 0.18$ and $\rho = 0.02$: fifty such claims carry the information of one grade. The process exit code does barely better ($\gamma = 0.25$, $\rho = 0.04$). An independent rating of 4 or 5 on a five-point scale reaches $\gamma = 0.72$, $\rho = 0.42$.

The relation between a rating and a defect is not monotone. At ratings 2 and 3 the chance of at least one major finding is 1.00 and 0.81; at rating 1 it is 0.12. The reason is visible in the run records: 92% of rating-1 runs produced nothing at all — they hit a cap or a provider error with no artefact. A run that does nothing has nothing to be wrong about. Any reward monotone in the rating would pay for doing nothing, and any reward monotone in the *absence* of findings would pay for it too. Definition 2 does not, because an empty artefact fails the tests the brief asked for.

Therefore the scoreboard splits. There is one Bernoulli metric, acceptance, whose cells carry a source dimension in $\mathcal{G}$ and whose per-source (a, b) are fitted alongside any rater severities; and a separate continuous metric on its own scale. The two are never summed into one cell. This is the resolution of G3.

5 The decision: one dial in dollars

Definition 3 (The objective). For a task with context x and a stake $L \geq 0$,

$$J(a; x, L) = \mathbb{E}[B(a, x)] + L(1 - P(g=1 | a, x)), \quad (5)$$

and the pick is $a^*(x, L) = \arg \min_{a \in \mathcal{A}} J(a; x, L)$.

L is the whole user-facing interface: the number of dollars a task that has to be redone, or that ships a defect, costs this practitioner. Latency enters, when it is wanted, as $J + \lambda \mathbb{E}[t(a, x)]$ with λ dollars per minute; every statement below survives that addition unchanged, since it only adds a second deterministic term to the bill.

Write $b(a) = \mathbb{E}[B(a, x)]$ and $f(a) = 1 - P(g=1 | a, x)$ at a fixed x , and let $P = \{(b(a), f(a)) : a \in \mathcal{A}\} \subset \mathbb{R}_{\geq 0} \times [0, 1]$ be the achievable set.

Proposition 2 (What the dial can reach). *a minimises $J(\cdot; x, L)$ for some $L \in [0, \infty)$ if and only if $(b(a), f(a))$ is an exposed point of $\text{conv}(P) + \mathbb{R}_{\geq 0}^2$ in a direction $(1, L)$ — equivalently, a vertex of the lower-left convex hull of P . Every such a is Pareto-optimal; the converse fails.*

Proof. $J(a; x, L) = \langle (1, L), (b(a), f(a)) \rangle$ is a linear functional with strictly positive first and non-negative second coefficient. A point of a finite set minimises a linear functional with non-negative coefficients over the set iff it minimises it over the convex hull, and the minimisers of such a functional over a polytope are exactly the faces exposed by that direction; ranging L over $[0, \infty)$ ranges the direction over all of $\{(1, L) : L \geq 0\}$, whose exposed points are the vertices of the lower-left hull. That a minimiser is Pareto-optimal for $L > 0$ follows because a dominating point would attain a strictly smaller value. The converse fails already for three points: with $P = \{(0.1, 0.40), (0.5, 0.30), (1.0, 0.10)\}$ the middle point is Pareto-optimal but lies strictly above the segment joining the other two (which passes through

$(0.5, 0.26)$), so no $(1, L)$ exposes it. This is the standard phenomenon of *unsupported* efficient solutions in bi-objective combinatorial optimisation [11, 9]. $\square$

Proposition 3 (The sweep is a finite monotone path). *Order the lower-hull vertices by increasing bill, $a_0, \dots, a_K$. Then $L \mapsto b(a^*(L))$ is non-decreasing and $L \mapsto f(a^*(L))$ is non-increasing; $a^*(L) = a_j$ exactly for L in the interval $[L_j, L_{j+1}]$ with*

$$L_j = \frac{b(a_j) - b(a_{j-1})}{f(a_{j-1}) - f(a_j)}, \quad L_0 = 0, \quad L_{K+1} = \infty, \quad (6)$$

and $L_1 \leq L_2 \leq \dots \leq L_K$. Each L_j is the dollars per unit of failure probability that the j th upgrade buys.

Proof. Consecutive hull vertices a_{j-1}, a_j are indifferent exactly when $b(a_{j-1}) + Lf(a_{j-1}) = b(a_j) + Lf(a_j)$, which rearranges to (6); the hull’s edge slopes $-1/L_j$ increase along the hull by convexity, so the L_j increase, and the argmin steps from a_{j-1} to a_j at L_j and nowhere else. Monotonicity of b and f along the sweep is then monotonicity along the hull, which holds by construction. $\square$

Proposition 3 is what makes the dial implementable: the hull is computed once, in $O(|\mathcal{A}| \log |\mathcal{A}|)$ after the front, and the answer for any L is a binary search over at most $K + 1$ intervals. It is also what makes the dial explainable: L_j is a sentence — “spending \$0.90 more per task buys one point of acceptance”.

5.1 Retiring the knee, and why the usual argument for retiring it is wrong

One expects to show that the knee of Eq. (2) is unreachable. It is not.

Proposition 4 (The knee is always supported). *Let the front have at least two distinct qualities, so the chord’s slope $g = (Q_1 - Q_0)/(\ln B_1 - \ln B_0)$ is strictly positive, and let a_p maximise the bracket of Eq. (2) over the front. Then $(b(a_p), f(a_p))$ is a vertex of the lower-left convex hull of P . Consequently the knee is the argmin of $J(\cdot; L)$ for a non-empty interval of L .*

Proof. Write $D(a) = Q(a) - Q_0 - g(\ln B(a) - \ln B_0)$, so $D(a_p) \geq D(a)$ for all a on the front, and recall $Q = 100(1 - f)$ so that “above the chord in Q ” and “below the chord in f ” are the same statement. Suppose a_p is not a hull vertex. Then there are front points a_u, a_v with $B_u < B_p < B_v$ such that a_p lies strictly above the segment $\overline{a_u a_v}$ in (B, f) , equivalently strictly below it in (B, Q) :

$$Q_p < Q_u + t(Q_v - Q_u), \quad t = \frac{B_p - B_u}{B_v - B_u} \in (0, 1),$$

i.e. $(1 - t)(Q_p - Q_u) < t(Q_v - Q_p)$. Optimality of a_p gives $D(a_p) \geq D(a_u)$ and $D(a_p) \geq D(a_v)$, that is

$$Q_p - Q_u \geq g \ln(B_p/B_u), \quad Q_v - Q_p \leq g \ln(B_v/B_p).$$

Combining, $(1 - t)g \ln(B_p/B_u) \leq (1 - t)(Q_p - Q_u) < t(Q_v - Q_p) \leq t g \ln(B_v/B_p)$, and since $g > 0$,

$$(1 - t) \ln(B_p/B_u) < t \ln(B_v/B_p). \quad (7)$$

But $B_p = (1 - t)B_u + tB_v$, so by concavity of $\ln$, $\ln B_p \geq (1 - t) \ln B_u + t \ln B_v$, which rearranges to exactly $(1 - t) \ln(B_p/B_u) \geq t \ln(B_v/B_p)$, contradicting (7). Hence a_p is a hull vertex, and by Proposition 2 some $L \geq 0$ selects it. $\square$

Remark 1. The proof uses only that the knee’s scalarisation is affine in a *concave* increasing transform of the bill. Any such transform — $\ln$, $\sqrt{\cdot}$, a Box–Cox with $\lambda \leq 1$ — gives the same conclusion. A *convex* transform would not, and a knee measured on a convex transform of the bill can indeed be unsupported.

So the knee is a decision. The objection is a different one, and it is fatal.

Proposition 5 (The knee’s stake is covariant with the price scale; the practitioner’s is not). *Multiply every price in the catalog by $\sigma > 0$, so $b \mapsto \sigma b$ and f is unchanged. Then (i) the knee of Eq. (2) is unchanged; (ii) $\arg \min_a \{\sigma b(a) + Lf(a)\} = \arg \min_a \{b(a) + (L/\sigma)f(a)\}$, so the assignment that a fixed stake L selects is the assignment that the stake L/σ selected before. If the lower hull has at least two vertices, there is an open interval of σ on which the two disagree.*

Proof. (i) $\ln(\sigma B) - \ln(\sigma B_0) = \ln B - \ln B_0$, and likewise for the denominator, so the bracket of (2) is unchanged. (ii) $\sigma b + Lf = \sigma(b + (L/\sigma)f)$ and multiplication by $\sigma > 0$ preserves the argmin. By Proposition 3, $L \mapsto a^*(L)$ is a non-constant step function whenever $K \geq 1$, so $a^*(L/\sigma) \neq a^*(L)$ for σ in an interval around any breakpoint ratio. $\square$

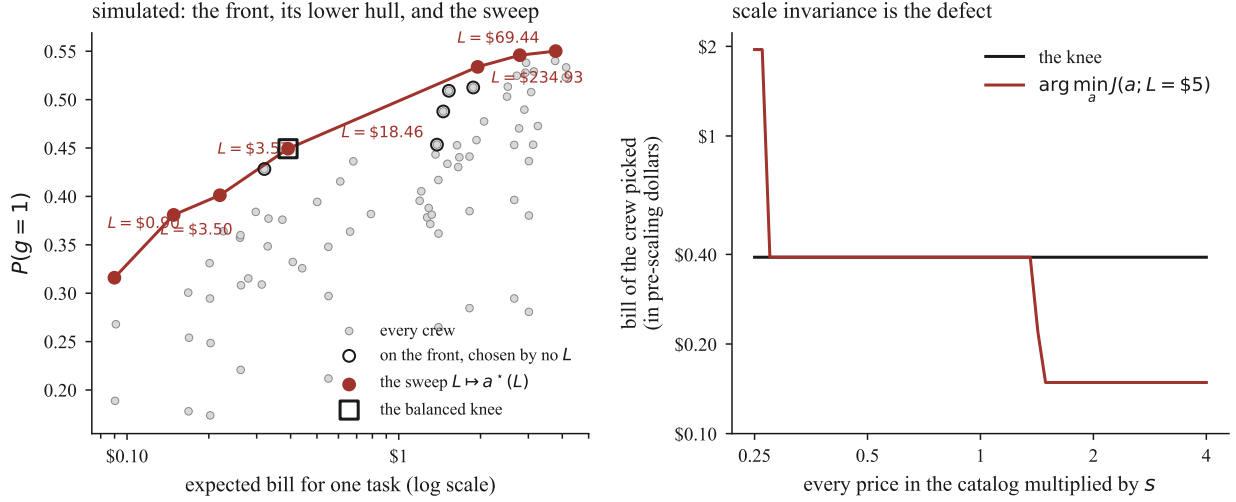


Figure 2: Simulated. Nine candidate models at each of two roles, prices log-uniform over the span the measured ledger shows, quality additive on the logit scale (Assumption 1). Left: all 81 assignments; the 12 on the Pareto front; the 7 lower-hull vertices, which are exactly the assignments some L selects (Proposition 2), with the breakpoints of Eq. (6); the 5 front assignments that no L selects; and the knee of Eq. (2), which is a hull vertex, as Proposition 4 requires. Right: every price multiplied by σ . The knee does not move. The assignment that a stake of \$5 selects moves twice.

This is the whole case. Prices in this market move by factors of two in months. Under a uniform halving of prices the correct decision moves *up* the front, because a fixed number of dollars of stake now buys twice as much quality; the knee does not move at all, because scale-invariance is what it was designed for. The right panel of Figure 2 draws that: the knee is a horizontal line, and the $L = \$5$ pick is a staircase crossing it. Worse, the stake the knee *implies* is never shown: on the simulated front of Figure 2 the knee sits in the interval $L \in [\$3.57, \$18.46]$, a range in which the practitioner was never asked to have an opinion.

6 Price bands

Definition 4 (Band). Fix a base $\beta > 1$ and a reference price $\pi_{\min}$. Model m ’s band at role r is $\text{band}_r(m) = \lfloor \log_\beta (c_{r,m}/\pi_{\min}) \rfloor$, where $c_{r,m}$ is the role’s expected bill for one task under the token shape of Section 8. We use $\beta = 2$: a band is a doubling of the bill.

Bands do two jobs and neither of them is the decision.

Pruning. $|\mathcal{A}|$ is a product over roles. With hundreds of priced models the full assignment set is far too large to score, and a quality floor — the usual device — can delete a whole price region. A band-aware shortlist keeps at most k models per band per role, the k with the largest posterior mean of $\theta_{r,m}$, which bounds $|\mathcal{A}|$ at $(k \cdot \#\text{bands})^{|\mathcal{R}|}$ and cannot delete a price region. We use $k = 3$. The rule is a heuristic and we do not claim it preserves the hull; an assignment whose only virtue is that it is the fourth-best in its band is lost.

Pooling exploration. A model with no observations borrows, through the hierarchy of Section 7, from its band: $\theta_{r,m} \sim \mathcal{N}(w_r^\top \varphi_m + \mu_{r,\text{band}(m)}, \sigma_0^2)$ with a band offset $\mu_{r,b}$ itself drawn from $\mathcal{N}(0, \sigma_b^2)$. Because bills within a band differ by less than a factor of β , the exploration budget of Section 9 can be spent on any member of a band at a known worst-case cost, which is what makes a hard dollar cap enforceable in advance rather than in arrears.

What bands are not. They are not the decision. A band is a bin over one coordinate of P ; the decision is Proposition 2. Reporting a band instead of a stake would reintroduce exactly the unit-free knob just retired.

7 Quality: per-role decomposition, prior, sample cost, drift

Assumption 1 (Additive logit).

$$\text{logit } P(g=1 \mid a, x) = \alpha_x + \sum_{r \in \mathcal{R}} \theta_{r,a(r)}. \quad (8)$$

What it buys. The unknowns are $\sum_r |\mathcal{M}_r|$ numbers rather than $|\mathcal{A}|$. Feedback is *semi-bandit*: the grade of one run is one observation for every role on that assignment, so a run informs every assignment that shares a model with

it. This is the structure of combinatorial bandits with semi-bandit feedback [2, 26], and it is the difference between learning tens of numbers and learning tens of thousands.

What it ignores. Eq. (8) has no interaction. The interaction one most expects is real and has a name: a strong reviewer rescuing a weak writer, so that $P(g=1)$ for (cheap writer, dear reviewer) exceeds the additive prediction. We do not fit it as a parameter. We capture the largest part of it structurally, by the cascade of Section 10, which is an explicit two-stage action whose value is computed rather than learned. A pair term $\delta_{(m_r, m_{r'})} \sim \mathcal{N}(0, \tau_{rr'}^2)$ can be added as a further level of the hierarchy below and costs observations only where a pair is actually run. We flag any claim about its size as conjecture — and Section 13 reports a hindsight result that the additive model gets wrong for exactly this reason, which is the strongest evidence we have that the term is worth fitting.

7.1 The prior, and the three-source posterior

Assumption 2 (Hierarchy). With φ_m the catalog indexes and w_r the role’s weighting,

$$\theta_{r,m} \sim \mathcal{N}(w_r^\top \varphi_m + \mu_{r,\text{band}(m)}, \sigma_0^2), \quad \theta_{r,m}^{(i)} \mid \theta_{r,m} \sim \mathcal{N}(\theta_{r,m}, \sigma_u^2), \quad (9)$$

where $\theta^{(i)}$ is site i ’s own value and σ_u^2 is the between-site variance. Observations carry variance σ^2 .

Every priced model has a prior mean on the day it appears, which is the cold-start property the catalog is for.

Proposition 6 (Three-source posterior). *Under Assumption 2, with $k = \sigma^2/\sigma_0^2$, $\ell = \sigma^2/\sigma_u^2$, $\tilde{n} = n^{\text{pool}}\ell/(\ell+1)$ and $K = k + \tilde{n}$, the posterior of site i ’s $\theta_{r,m}^{(i)}$ given the catalog, the pool and n^i own observations is normal with*

$$\mathbb{E}[\theta_{r,m}^{(i)} \mid \cdot] = \frac{k^* \hat{\theta}^{\text{cat}} + n^* \bar{y}^{\text{pool}} + n^i \bar{y}^i}{k^* + n^* + n^i}, \quad \text{Var}[\theta_{r,m}^{(i)} \mid \cdot] = \frac{\sigma^2}{k^* + n^* + n^i}, \quad (10)$$

where everything external to this site shares one capped count

$$W = \frac{K\ell}{K+\ell} \leq \ell, \quad k^* = \frac{k}{K}W, \quad n^* = \frac{\tilde{n}}{K}W. \quad (11)$$

Proof. Integrate out $\theta_{r,m}$. The catalog and the pool inform $\theta_{r,m}^{(i)}$ only through $\theta_{r,m}$, whose posterior given them is normal with precision K/σ^2 and mean the $k : \tilde{n}$ precision-weighted average of $\hat{\theta}^{\text{cat}}$ and $\bar{y}^{\text{pool}}$; each pool observation is another site’s, hence carries variance $\sigma^2 + \sigma_u^2$ about $\theta_{r,m}$, which is the deflation $\tilde{n} = n^{\text{pool}}\ell/(\ell+1)$. Adding the between-site variance σ_u^2 gives, for $\theta^{(i)}$, precision $1/(\sigma^2/K + \sigma_u^2) = W/\sigma^2$ with the same mean. Own observations add precision n^i/σ^2 directly. Precision-weighting the two gives (10), and $k^* + n^* = W$ split in the ratio $k : \tilde{n}$. $\square$

Corollary 1. *With $\sigma_u = 0$ ($\ell = \infty$) and no band offset, Eq. (10) is the plain shrinkage blend $(1-w)\hat{\theta}^{\text{cat}} + w\bar{y}$ with $w = n/(n+k)$, $n = n^{\text{pool}} + n^i$: exactly Eq. (3), with $\kappa = k = 30$. Baseline B1 is therefore the degenerate case of Proposition 6 in which every other site’s run is assumed to be as informative as one’s own.*

Eq. (11) is the point of the hierarchy: however many rows the pool holds, everything external to this site is worth at most $\ell = \sigma^2/\sigma_u^2$ of this site’s own observations, because other sites’ tasks are not this site’s tasks; own observations are never capped. Section 12 gives ℓ its estimator.

7.2 How many runs it costs to tell two models apart

Proposition 7 (Sample complexity at one role). *Let two models at role r differ by Δ in $P(g=1)$ at a common operating point $\bar{p}$, let each have κ prior observations and N graded ones, and approximate the grade’s variance by $\bar{p}(1-\bar{p})$. Under the conjugate normal approximation the posterior means differ by a normal with variance $2\bar{p}(1-\bar{p})/(\kappa+N)$, so a two-sided test at level z separates them when*

$$N \geq \frac{2z^2 \bar{p}(1-\bar{p})}{\Delta^2} - \kappa. \quad (12)$$

Proof. Immediate from the conjugate posterior $\mathcal{N}(\cdot, \sigma^2/(\kappa+N))$ with $\sigma^2 = \bar{p}(1-\bar{p})$, independence of the two arms, and the definition of a z -level two-sided test. $\square$

At $z = 1.96$ and $\bar{p} = 0.75$ (so $\bar{p}(1-\bar{p}) = 0.1875$), Eq. (12) gives, per model, at one role:

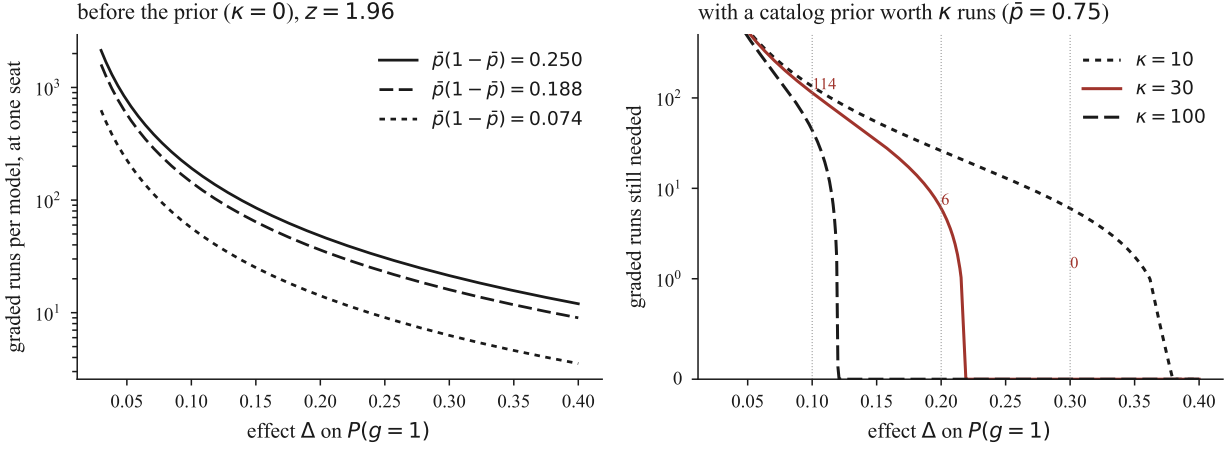


Figure 3: Closed form. Eq. (12). Left: graded runs per model at one role before any prior, for three operating points; $\bar{p}(1 - \bar{p}) = 0.074$ is the major-free rate the measured ledger shows, 0.25 the worst case. Right: what remains after a catalog prior worth κ runs, at $\bar{p} = 0.75$, with the $\kappa = 30$ of baseline B1 marked. Remark 2 applies to every curve on the right.

Δ	$\kappa = 0$	$\kappa = 10$	$\kappa = 30$	$\kappa = 100$
0.10	144	134	114	44
0.20	36	26	6	0
0.30	16	6	0	0

Remark 2 (The subtraction is a loan, not a gift). The $-\kappa$ in (12) is only a saving if the prior is centred correctly. A prior worth κ observations that is wrong by more than Δ does not shorten the experiment; it lengthens it, because the first κ observations are spent undoing the prior. Baseline B1’s $\kappa = 30$ is a constant, not an estimate; Section 12 gives the moment estimator that should replace it, and Section 9 shows what an overconfident κ costs in the one regime where it matters most.

Two corollaries. First, $\Delta = 0.10$ is out of reach for a single site: 114 graded runs *per model per role*, against 202 runs in total across all models in the ledger of Section 13. Pooling is not an optimisation; it is the only way the question is answerable at all. Second, semi-bandit feedback is what makes even $\Delta = 0.20$ plausible, because each run pays for $|\mathcal{R}|$ of these counters at once.

7.3 Drift

Models change under one identifier. Between observations, inflate the variance:

$$\theta_{r,m,t} = \theta_{r,m,t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \tau^2 \Delta t), \quad V_t = \min(\sigma_0^2, V_{t-1} + \tau^2 \Delta t), \quad (13)$$

which is the state equation of a local-level model [13] and the variance-inflation step of Glicko-2 [12]; the cap at the prior variance is what keeps an arm from becoming *more* uncertain than a model nobody has ever run. Two consequences. An arm nobody has run for a while regains uncertainty and is drawn again by the sampler of Section 9 — forgetting is what makes exploration self-renewing without an exploration rate. And a model identifier that is re-released is a change point: the mean and the variance both reset to the prior, rather than drifting there. Figure 4 draws both. τ^2 is estimated by the day-to-day moment $\hat{\tau}^2 = [\widehat{\text{Var}}(\bar{y}_d - \bar{y}_{d-1}) - \sigma^2(1/n_d + 1/n_{d-1})]_+$ pooled over cells, which is why the day is a cell dimension; a cell with no evidence of drift gets $\hat{\tau}^2 = 0$ and forgets nothing. The non-stationary bandit literature [10] offers discounted and sliding-window alternatives; we prefer variance inflation because it leaves the posterior mean alone and therefore does not throw away information that has not been contradicted.

8 Cost: a learned token shape

Definition 5 (Token shape). $T_{r,m} = (\bar{l}_{r,m}, \bar{y}_{r,m}, \bar{o}_{r,m}, \bar{v}_{r,m})$: mean input tokens per call, the fraction of them served from cache, mean output tokens per call, and mean calls per task, for model m in role r .

The bill is then

$$\mathbb{E}[B(a, x)] = \sum_{r \in \mathcal{R}} \bar{v}_{r,a(r)}(x) \left[(1 - \bar{\gamma}) \bar{l} \pi_{a(r)} + \bar{\gamma} \bar{l} \pi_{a(r)}^c + \bar{o} \chi_{a(r)} \right], \quad (14)$$

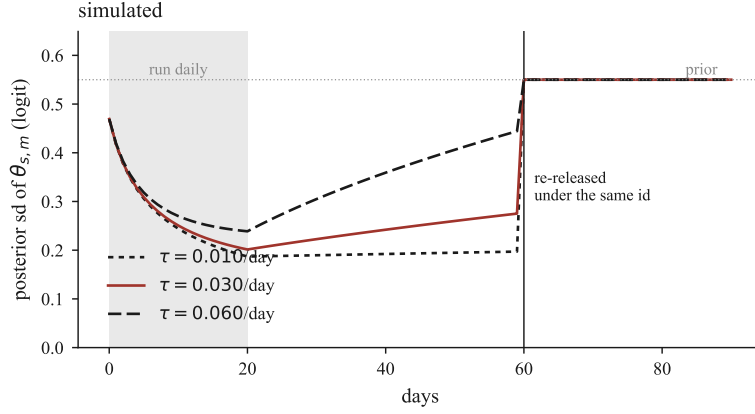


Figure 4: Simulated. Eq. (13). An arm run daily for three weeks, then never; its posterior standard deviation climbs back toward the prior at a rate set by τ , which is what makes it eligible to be drawn again. At day 60 the identifier is re-released and the arm is reset rather than drifted.

with the four shape components read at the task’s context x . Each is estimated from the site’s own usage ledger with a shrinkage prior at the role’s default shape:

$$\hat{T}_{r,m} = \frac{n_{r,m} \bar{T}_{r,m} + \kappa_T T_r^{\text{def}}}{n_{r,m} + \kappa_T}, \quad (15)$$

a model with no history priced exactly as the baseline prices it, a model with history priced as it actually behaves.

The shrinkage is on the log scale, and this is not a detail. Eq. (15) is written as an arithmetic average, which is what a conjugate normal model prescribes for a quantity whose errors are additive. The volume share of a role is not such a quantity: it is bounded in $[0, 1]$, heavy-tailed, and enters the bill multiplicatively. Section 13 measures the difference and it is the entire effect: shrinking the arithmetic mean of the role’s volume share improves leave-one-out bill prediction from a mean absolute percentage error of 302% to 242%, while shrinking its logarithm — that is, replacing the arithmetic mean in (15) by a geometric one for the volume components — improves it to 58%. The reason is visible in Figure 7: a reviewing role takes a few per cent of a run’s tokens on most runs and most of them on a few, so the arithmetic mean of its share is not the share of any typical run, and a rule that prices the typical run from that mean is wrong on nearly every run.

The expectation is over task volumes, not at the median task. J takes an expectation, so the cost model must integrate (14) over the distribution of task volumes rather than evaluate it at a representative task. The distribution is heavy-tailed and its mean is not its median; Section 13 shows that pricing the median task understates the mean bill of an assignment by a factor that varies across assignments, which reorders them.

9 Exploration

The pick under Eq. (5) at posterior means never tries an assignment the prior underrates. Thompson sampling [24, 21] draws one plausible world and acts as if it were true.

Four things in Algorithm 1 deserve their own justification.

The draw is inside J , not beside it. Because costs are deterministic and L is in dollars, an expensive assignment is drawn only when the sampled quality gain covers its extra bill at the practitioner’s own stake. There is no separate exploration rate to set and no unit-free temperature.

A hard daily cap E . Thompson sampling has no budget; J prices exploration but does not bound it. E is a per-site, per-day cap on $\sum_t [B(\tilde{a}_t) - B(\hat{a}_t)]_+$, which makes the worst case a number the practitioner can be told. This is the bandits-with-knapsacks setting [1] in its simplest form — one resource, a hard constraint — and we do not use its algorithms; we use the constraint, and we say plainly that greedily spending the budget until it is exhausted is not the optimal way to spend it.

One role changes per task. Changing two roles at once makes the resulting grade uninformative about which change caused it, and makes bills jump. The restriction costs optimality — the reachable set in one step is a neighbourhood, not $\mathcal{A}$ — and buys attribution and stability.

Algorithm 1 One pick

Require: task context x ; stake L ; per-role shortlists M_r (at most k per band, Section 6); the day’s remaining exploration budget E_{left} ; the incumbent assignment a_0

- 1: **for** each role r and $m \in M_r$ **do**
- 2: $\tilde{\theta}_{r,m} \sim \mathcal{N}(\mu_{r,m}, V_{r,m})$ ▷ one draw per task; never redrawn inside a task
- 3: **end for**
- 4: $\tilde{a} \leftarrow \arg \min_a \mathbb{E}[B(a, x)] + L(1 - \text{logit}^{-1}(\alpha_x + \sum_r \tilde{\theta}_{r,a(r)}))$ ▷ costs are *not* sampled
- 5: $\hat{a} \leftarrow$ the same argmin at the posterior means μ ▷ the greedy assignment
- 6: $e \leftarrow [\mathbb{E}[B(\tilde{a}, x)] - \mathbb{E}[B(\hat{a}, x)]]_+$ ▷ what this draw would spend on exploration
- 7: **if** $e > E_{\text{left}}$ **then**
- 8: $\tilde{a} \leftarrow \hat{a}; e \leftarrow 0$ ▷ the hard daily cap
- 9: **end if**
- 10: $\tilde{a} \leftarrow$ the assignment in $\{a_0\} \cup \{a : a \text{ differs from } a_0 \text{ in at most one role}\}$ nearest $\tilde{a}$ in J ▷ one role moves per task
- 11: **return** $\tilde{a}$ if $J(a_0) - J(\tilde{a}) > \delta$ (Section 11), else a_0 ; $E_{\text{left}} \leftarrow E_{\text{left}} - e$

Cheap exploration has its own vehicle. A one-token probe action buys a measurement at negligible cost, and reserving a slot for the least-observed pair is the same idea at the task level. Observations from a deliberately exploratory slot carry a source tag and hence, by Proposition 1, their own reliability; they are not the same evidence as a run the rule chose on its merits.

9.1 What the simulation says, including the part we did not want

Figure 5 is a calibrated simulation: two roles, ten models each, bills log-uniform over the two orders of magnitude the measured ledger spans, quality additive on the probability scale, grades Bernoulli, estimation the exact Gaussian linear model (so one grade updates both roles — the semi-bandit structure), prior the truth plus noise held at strength κ .

Where every model carries a catalog index, exploration does not repay itself. Over 800 tasks the posterior-mean rule accumulates \$195 of regret, Thompson inside J accumulates \$347, and the capped version \$270. Thompson’s instantaneous regret falls below the greedy rule’s only at around $T \approx 2000$ in a longer run, and the cumulative debt is not repaid within any horizon a single site will see. The reason is not subtle: a Bernoulli grade has standard deviation near 0.45 about a quantity whose spread across models is near 0.15, so evidence arrives slowly, while $L = \$5$ makes a quality uncertainty of 0.04 worth \$0.20 — more than the whole bill of a cheap assignment. We checked the sensitivity: at $L \in \{\$1, \$5, \$20, \$80\}$ the ratio of Thompson’s regret to the greedy rule’s is 1.36, 1.51, 1.70, 1.73. The finding is robust in the direction we did not want.

Where one model per role carries no index, the greedy rule fails completely. Give each role one model the catalog does not index — a new release, an open-weights checkpoint, anything outside the feed — give it a prior equal to the role’s average with three times the prior variance, and make it in fact the best model at its role. The posterior-mean rule never plays it: its regret over 800 tasks is \$839 and its per-task regret in the last hundred tasks is \$1.00, not falling. Thompson finds it: \$301 cumulative, \$0.19 per task and still falling. The cap costs almost nothing here (\$273 cumulative regret, \$4 of exploration spend against \$70 uncapped) because the exploration that matters is cheap and early.

The design conclusion. Exploration is not worth paying for in the assignment choice at this data scale when the prior covers every arm; it is worth almost any price when the prior does not cover an arm. So: keep the posterior-mean pick as the default, keep the Thompson draw behind a setting, keep the hard cap E small, and direct the budget at arms whose *prior* is uninformative rather than at arms whose posterior is merely wide. That is a value-of-information argument, and it is the argument for IDS [19, 20] or a knowledge-gradient-per-dollar rule over plain Thompson sampling. We do not implement either, for a stated reason: both need a per-candidate information estimate at every pick, and at $|\mathcal{A}|$ in the thousands and $|\mathcal{M}|$ in the hundreds that is a second optimisation inside the first, while the gain over Thompson sampling at T in the hundreds is within the noise of Figure 5. We regard “IDS beats Thompson-in- J on this problem” as a conjecture and have not tested it.

9.2 Regret, as a sketch

Proposition 8 (Sketch). *Suppose (i) $P(g=1 | a, x)$ is replaced by its linearisation $v^\top \theta_a$ about the operating point, so that $J(a; L) = b(a) + L(1 - v^\top \theta_a)$ is affine in θ with a known deterministic offset $b(a)$; (ii) the prior is the true generating prior; (iii) grades are conditionally sub-Gaussian with proxy ς . Then the Bayesian regret of Thompson*

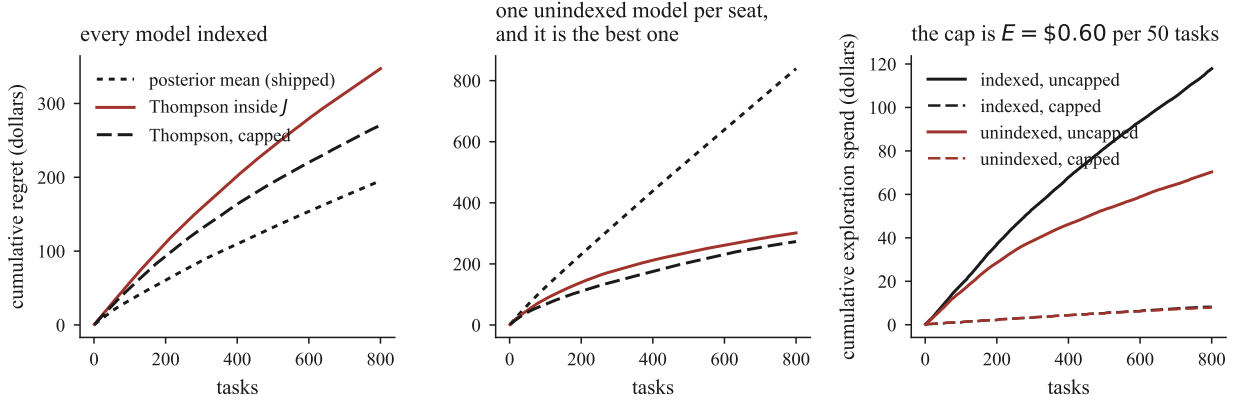


Figure 5: Simulated, calibrated to the measured ledger’s cost spread. Left: with every model indexed, the posterior-mean rule beats Thompson sampling inside J over any horizon a site will see. Middle: with one unindexed model per role that happens to be the best, the posterior-mean rule’s regret does not fall at all. Right: what exploration costs, and what the per-day cap holds it to. The conclusion is the asymmetry, not either panel alone.

sampling over T tasks satisfies $\mathcal{R}(T) = O(L \varsigma \sqrt{dT \log T})$ with $d = \sum_r |M_r|$, by the general posterior-sampling bound of Russo and Van Roy [18], and the d rather than $|\mathcal{A}|$ is exactly the semi-bandit structure exploited by the combinatorial analyses of Chen, Wang and Yuan [2] and Wen, Kveton and Ashkan [26]; for combinatorial Thompson sampling specifically see Wang and Chen [25].

We label this a sketch and mean it. Assumption (i) is false — the link is a logit and the linearisation error is not controlled; assumption (ii) is false, and Section 9 showed the failure of (ii) is what actually governs the outcome; the band pruning of Section 6 changes $\mathcal{A}$ between rounds, which none of these analyses covers; and the exploration cap makes the policy non-Thompson on the rounds where it binds. The bound is offered as an account of *why* the dimension is d and not $|\mathcal{A}|$, and for nothing else. The empirical statement in Figure 5 is not an instance of it.

10 Cascades

Extend the action set. A *cascade* ($b \rightarrow b'$) runs an assignment from band b ; if the grade is 0, it escalates to an assignment from band $b' > b$; if that also fails, the stake is paid.

Proposition 9 (Break-even). Write $c_b, c_{b'}$ for the two bands’ expected bills and $p_b, p_{b'}$ for their acceptance probabilities, and let $X = c_{b'} + (1 - p_{b'})L$ be the cost of going straight to b' . Then

$$J(b \rightarrow b') = c_b + (1 - p_b)[c_{b'} + (1 - p_{b'})L], \quad (16)$$

$$J(b \rightarrow b') < J(b') \iff p_b > \frac{c_b}{c_{b'} + (1 - p_{b'})L}, \quad (17)$$

$$J(b \rightarrow b') < J(b) \iff p_{b'}L > c_{b'}. \quad (18)$$

At $L = 0$, (17) is the bare cost ratio $p_b > c_b/c_{b'}$.

Proof. (16) is the expectation over the first grade. Then $J(b \rightarrow b') - J(b') = c_b + (1 - p_b)X - X = c_b - p_bX$, giving (17). And $J(b \rightarrow b') - J(b) = (1 - p_b)[c_{b'} + (1 - p_{b'})L] - (1 - p_b)L = (1 - p_b)[c_{b'} - p_{b'}L]$, which is negative iff $p_{b'}L > c_{b'}$, giving (18). With a latency price λ the same algebra holds with c replaced by $c + \lambda t$ throughout. $\square$

Read together the two conditions say: cascade when the cheap band clears the bar often enough to pay for its own attempt, *and* when the escalation is worth buying at the practitioner’s stake at all. Higher stakes lower the first bar and raise the second, because a cascade ends unaccepted with probability $(1 - p_b)(1 - p_{b'})$, strictly less than $(1 - p_{b'})$: escalation buys quality as well as costing money. Figure 6 draws both. This is the mechanism FrugalGPT studies for single-call inference [3]; the difference here is that the escalation decision is made on a model-free grade rather than on a scored confidence, which is what the next paragraph is about.

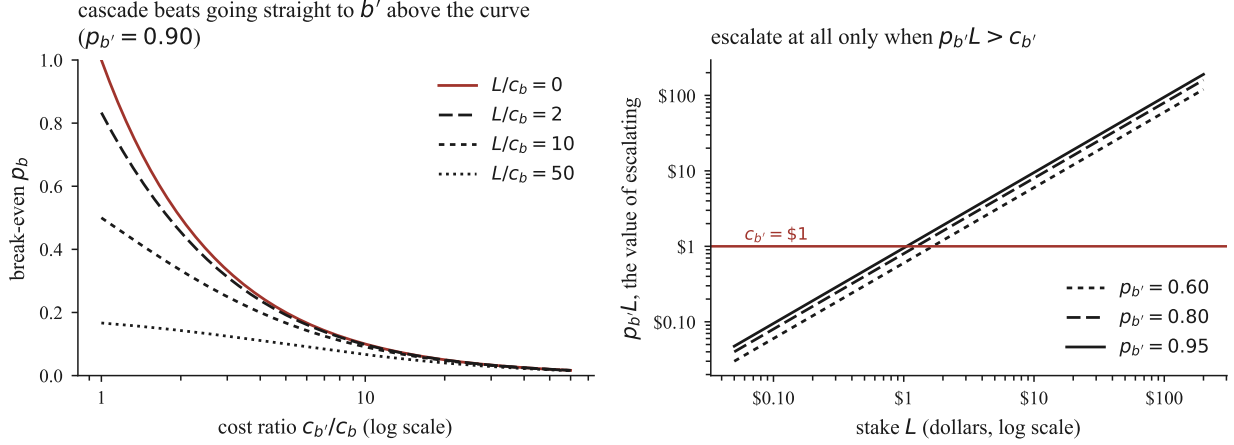


Figure 6: Closed form. Proposition 9. Left: the break-even acceptance for the cheap band against the cost ratio, for four stakes; $L = 0$ is the bare cost ratio, and every dollar of stake lowers the bar because a cascade strictly reduces the chance of an unaccepted task. Right: condition (18), which decides whether the escalation is worth buying at all.

The endogeneity hazard, and why an executable grade removes it. Suppose the escalation trigger were the reviewing role’s verdict rather than the grade, and suppose that role is part of the assignment being scored. Then a cheap reviewer that passes bad work produces no escalation, the cheap writer’s measured acceptance rises, the rule prefers the cheap assignment, and the cheap reviewer is chosen again. The feedback loop is closed and it collapses to the cheapest possible assignment with the worst possible quality, while every measured quantity in the system improves. Formally, the trigger z and the outcome g are both functions of the action a through the reviewing role, so $\mathbb{E}[z | a]$ is not an unbiased signal of $\mathbb{E}[g | a]$ and the argmin of the estimated J is not consistent for the argmin of J . Definition 2 breaks the loop by construction: g is computed from the artefact by the harness, and no model in the assignment can move it. An independent rating and the reviewing role’s findings remain as auxiliary observations with reliability $\rho < 1$, which is exactly the right treatment for a signal that is informative and endogenous: it can shorten the experiment, and it cannot determine its answer.

11 Switching and stability

Definition 6 (Switching threshold). Let a_0 be the incumbent and a_1 the challenger. Switch only at a task boundary, and only when

$$J(a_0) - J(a_1) > \delta, \quad \delta = \underbrace{\hat{t}^{\text{cached}}(\pi_{a_0(r)} - \pi_{a_0(r)}^c)}_{\text{cache loss}} + h, \quad (19)$$

where $\hat{t}^{\text{cached}}$ is the prompt prefix currently cached for the role being changed and $h \geq 0$ is a hysteresis margin.

The first term is a real cost with a receipt: changing the model in a role discards the provider-side cache of that role’s prompt prefix, and the next call pays the prompt price for tokens that would have been served at the cache-read price. The second is a margin against churn.

Remark 3 ($\delta = 0$ for a batch harness). Where every task is a fresh session, there is no cached prefix to lose across tasks, so $\hat{t}^{\text{cached}} = 0$ and $\delta = h$. Inside an interactive session the prefix is real and $\delta > 0$; this is why the model in a role is fixed for the duration of a task and may change only between tasks.

Evaluating at every boundary inflates false switches. The rule of Eq. (19) is tested after every task, which is optional stopping: a fixed- α test applied at every t rejects with probability one under the null. The correct instrument is an anytime-valid confidence sequence: a family $\{C_t\}$ with $P(\exists t : \theta \notin C_t) \leq \alpha$, built from a non-negative supermartingale [15]. Concretely, maintain the sequence for the difference $J(a_0) - J(a_1)$ and switch when its lower bound exceeds δ . The cost is a wider interval than a fixed-sample test at the same α ; the benefit is that “we checked at every task” is no longer a way of manufacturing a switch.

Why batching is the right shape. Bandits with switching costs have minimax regret $\tilde{O}(T^{2/3})$ rather than $\tilde{O}(T^{1/2})$ [5], and the optimal policies batch: they hold an arm for a stretch, then reconsider. The task boundary is a natural batch

edge — it is where the cache is worthless anyway and where a bill is closed — so the switching structure the theory recommends is the structure the problem already has. We claim no bound; we claim that the shape is not accidental.

12 Pooling across sites

Definition 7 (Cell). A cell is keyed by (role, model, source, channel, size, day) and holds (n, Σ_1, Σ_2) : count, sum and sum of squares, plus the count of contributing sites. A sheet maps (site, cell) to such a triple.

Merging. Two structures, deliberately different. *Within* a store, a site’s own sheet accumulates its own disjoint observations by addition, which is associative and commutative but not idempotent, so a retried batch can be double-counted and the relay bounds that with a per-site daily quota. *Between* stores, a copy of another store’s sheet merges by componentwise maximum on each site’s running total, which is a join-semilattice and hence a state-based CRDT [22]: any two nodes that have seen the same publications, in any order and with any duplication, hold the same sheet. The published index is a function of the merged sheet, so a relay is a convenience for bandwidth and signing, not an authority.

What this is and is not. It is centralised-relay and mergeable: one relay publishes, any node can host one, and two relays that receive the same publications agree. It is *not* decentralised learning. There is no coordinated exploration — nothing tells site B that site A has already paid to measure a cell, so the pool re-measures what it already knows. Publishing a per-cell value of information alongside each cell, so that sites allocate their exploration budgets to disjoint questions, is the obvious next step and is exactly the distributed-exploration problem [14]; federated bandit formulations [23] give the communication–regret trade-offs. We have implemented none of it and claim none of it.

The site random effect gives ℓ its meaning. Assumption 2’s $\theta_{r,m}^{(i)} = \theta_{r,m} + u_{i,r,m}$, $u \sim \mathcal{N}(0, \sigma_u^2)$, is what makes $\ell = \sigma^2 / \sigma_u^2$ the cap of Eq. (11): however large the pool, the outside world is worth at most ℓ of this site’s own runs. Every constant is a moment estimate on the sheet, none is set by hand:

$$\begin{aligned} \hat{\sigma}^2 &= \frac{\sum_{i,\theta} (\Sigma_{2,i\theta} - \Sigma_{1,i\theta}^2 / n_{i\theta})}{\sum_{i,\theta} (n_{i\theta} - 1)}, & \hat{\sigma}_u^2 &= \left[\widehat{\text{Var}}_i(\bar{y}_{i\theta}) - \hat{\sigma}^2 \overline{(1/n_{i\theta})} \right]_+, \\ \hat{\sigma}_0^2 &= \left[\widehat{\text{Var}}_{(r,m)}(\bar{y}_{r,m}^{\text{pool}} - \hat{\theta}_{r,m}^{\text{cat}}) - (\hat{\sigma}^2 + \hat{\sigma}_u^2) / n^{\text{pool}} \right]_+, & \kappa &= \hat{\sigma}^2 / \hat{\sigma}_0^2, \quad \ell = \hat{\sigma}^2 / \hat{\sigma}_u^2, \end{aligned} \quad (20)$$

each a within-minus-noise estimator of the Efron–Morris kind [8]. Each is withheld until its denominator carries at least ten degrees of freedom, with the day-one values of Table 2 until then. On a small seed index $\hat{\sigma}_0^2$ clips to zero — a dozen cells of two-point scores cannot resolve a catalog error under about thirteen points — which is why $\kappa = 30$ still stands as a day-one value and why Remark 2 is not an academic caveat.

Robustness and reliability. Rows from one site share a task mix and a codebase; the unit of aggregation is therefore the site mean, not the row. A Huber M -estimate over site means (MAD scale 1.4826, tuning constant 1.345, breakdown 1/2) [16] bounds what a minority of broken or adversarial sites can move, with a precision-weighted mean below five sites. Rater severities are fitted as one additive effect per rater by alternating weighted means, which is the two-way additive model identified up to one constant per connected component of the rater-by-cell graph — the many-facet Rasch calibration [17]. The same machinery fits one $(a_{\text{src}}, b_{\text{src}})$ per source [4] and publishes ρ_{src} of Eq. (4) in the index, so that every site discounts a proxy observation by the same factor.

13 Experiments

13.1 Setup

The evidence is one production harness’s ledger over a 21-hour window, one repository, one language.¹ It holds 202 graded runs. Of these, 136 carry the full assignment (writer, reviewer, planner) together with an executable-proxy outcome, and 132 carry a per-role split of tokens and dollars. All model identifiers are as published on the day of the

¹The ledger comes from a production multi-role coding-agent harness operated by the authors. The harness itself is not the subject of this paper; every quantity below is a measurement of the variables defined in Section 2. The extraction script, the derived tables and every figure are released; the run records, which carry task text, are not.

constant	meaning	estimator	day-one value
σ	grade / score noise	(20)	— (fitted)
$\kappa = \sigma^2 / \sigma_0^2$	the catalog’s weight, in runs	(20)	30
$\ell = \sigma^2 / \sigma_u^2$	cap on outside evidence	(20)	30
τ^2	drift per day	moment on daily differences	0 (no forgetting)
ρ_{src}	a source’s reliability	(4), fitted per source	executable grader 1
β, k	band base; keep per band per role	fixed by design	2, 3
κ_T	shrinkage of the token shape	(15), on the log scale	30 runs
L	the stake, in dollars	the practitioner’s	\$1 / \$10 / ∞
λ	dollars per minute	the practitioner’s	\$0.01
E	daily exploration cap	the practitioner’s	a fraction of the daily bill
h	hysteresis in (19)	fixed by design	small

Table 2: Every constant with its estimator. Four belong to the practitioner by design; the rest are empirical-Bayes estimates the relay publishes with the index, or stated design judgements. No constant in this table is a unit-free knob, and only L is normally set.

catalog snapshot, which lists 522 models, 417 of them priced; prices are read from that snapshot and are per token. No run is dropped for a missing price: every model that appears in the ledger is priced in the snapshot.

Table 3 lists the eight models the ledger exercises, with their as-published prices and indexes.

model (as published)	\$/Mtok in	\$/Mtok out	index	roles held (runs)
anthropic/claude-fable-5.1	10.00	50.00	53.4	reviewer (43)
anthropic/claude-opus-5	5.00	25.00	50.7	planner (136), reviewer (3)
moonshotai/kimi-k3	3.00	15.00	43.8	reviewer (31)
qwen/qwen3.8-max-0902	2.00	6.00	45.4	reviewer (59)
z-ai/glm-5.3	1.40	4.40	44.9	writer (16)
meta/muse-spark-1.3	1.25	4.25	44.3*	writer (14)
deepseek/deepseek-v4.1-flash	0.15	0.60	39.5	writer (62)
z-ai/glm-5.3-flash	0.09	0.30	41.9	writer (44)

Table 3: The models the ledger exercises, as published in the catalog snapshot. Prices are quoted per million tokens for readability; the released table carries them per token, which is how the catalog states them. *imputed from the secondary index. The design is close to a balanced 4×4 over writer and reviewer with the planner held fixed, which is what makes the per-role marginals of Section 13 readable at this sample size.

Three quantities are fixed once here and used throughout. The *failure bit* is $1 - g$ with g taken to be the reviewing role’s major-defect flag, negated, and with an artefact that contains no change counted as a failure. This is a *proxy* for Definition 2, not the grade itself: the historical runs predate an executable grader. Section 4 measured what that proxy is worth in the other direction — an independent rating carries $\rho = 0.42$ of a grade — and every claim below inherits that qualification. The failure rate over the 136 graded runs is 0.176. The *assignment set* is the 12 distinct assignments run at least three times, out of 15 observed; the three dropped were run once each. The *quality model* is the MAP of Assumption 1 under a catalog-anchored Gaussian prior worth $\kappa = 30$ runs at the operating point, with per-role anchors $\lambda(\varphi_m/\varphi_{\max} - \bar{\varphi}_r)$ at $\lambda = 4$ logits; a model missing the primary index is imputed from the secondary one by least squares over the 58 catalog models carrying both ($r = 0.88$).

13.2 E1: what a role costs, and whether a learned shape predicts it

Figure 7, left, is the measured token shape. Over the 132 runs carrying a split, the writing role takes a median 0.974 of a run’s tokens (mean 0.877), the reviewing role a median 0.026 (mean 0.117) with a long tail — its ninetieth percentile is 0.395 and its maximum 0.946, and on nine of the 132 runs the role that only reads the work takes more than half the tokens — and the planning role a median below 10^{-3} (mean 0.006). The dollars land somewhere else entirely: the writing role takes a median 0.519 of them and the reviewing role a median 0.437. Restricted to the 116 runs in which the reviewing role was actually invoked, its token median is 0.034 and its dollar median 0.509. So the reviewing role’s bill share is mostly price and partly the volume tail. Against a fixed shape that assigns the reviewing role 0.08 of a run’s tokens, the measured mean is 1.46 times that (1.67 times on the invoked subset) while the measured median is a third of it — the fixed value is neither.

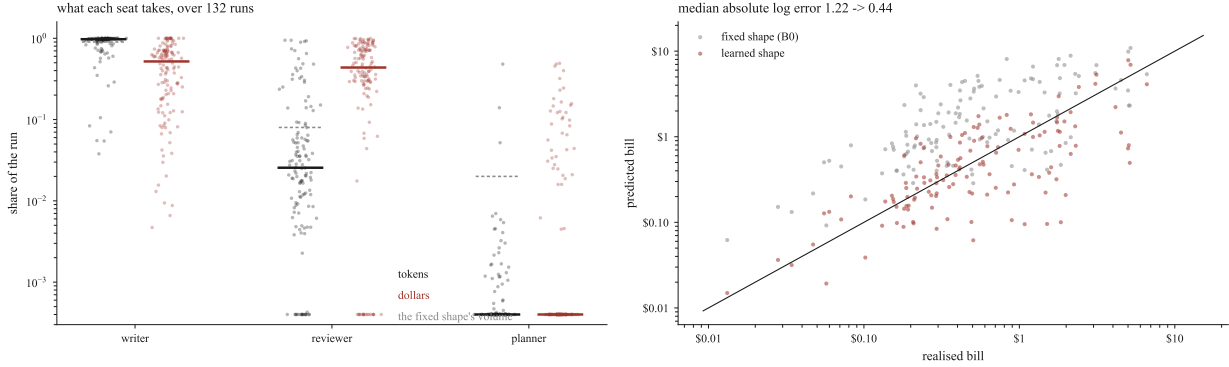


Figure 7: Measured. Left: each role’s share of a run, in tokens and in dollars, over the 132 runs that carry a per-role split; bars are medians, the dashed rules are the fixed shape’s volume shares. The writing role takes nearly all the tokens and about half the dollars; the reviewing role takes a few per cent of the tokens on a typical run, most of them on a few, and about half the dollars. Right: leave-one-out bill prediction, predicted against realised, for the fixed shape and for the learned shape with log-scale shrinkage. The fixed shape is biased high by a median factor of 3.3.

The prediction task is: given the assignment and the run’s total token volume, return the run’s bill. Leave-one-out throughout; the shape for a run is fitted on the other 131 runs only, shrunk toward the fixed shape at $\kappa_T = 30$; prices as published.

cost model	MAPE	median log error	median error factor
fixed shape (B0, B1)	302.5%	1.217	3.38× (over)
learned shape, arithmetic shrinkage	241.9%	1.058	2.72× (over)
learned shape, log-scale shrinkage	58.4%	0.443	0.92×

Two readings. First, the fixed shape is not merely noisy but biased: its median prediction is 3.3 times the realised bill, because it attributes 8% of a large token volume to a role whose price per token is often an order of magnitude higher than the writer’s, and on most runs that role does a small fraction of that work. Second, and this is the result we did not expect, *how* the shape is shrunk carries the entire effect. Shrinking the arithmetic mean of the volume share recovers a fifth of the gap; shrinking its logarithm recovers nine tenths of it. An ablation isolates the component: learning the volume share alone gives MAPE 56%, while learning the cache share alone gives 299% and the input-to-output split alone 305%. The bill of a multi-role run is, to a first approximation, a question about who spent the tokens.

At the assignment level — predicting the mean bill of each of the 12 assignments, which is what J needs — the learned shape gives MAPE 38.9% against 129.7% for the fixed one, and the fixed shape over-predicts by a median factor of 2.18. That over-prediction is not uniform across assignments, which is why it reorders them, and reordering is what E2 tests next.

13.3 E2: does the rule pick what hindsight would have picked?

For each stake $L \in \{\$1, \$10, \infty\}$ we compute, over the 12 assignments, the one with the lowest *realised* loss — mean bill plus L times failure rate, and for $L \rightarrow \infty$ the lowest failure rate with the bill as tie-break — and ask whether the rule’s pick, formed from the catalog prior plus the measured cells and priced by the learned shape, coincides with it. Figure 8 draws the realised set with Wilson intervals and marks both.

It does not coincide, at any of the three stakes, and the honest account of why is instructive.

At $L = \$1$ hindsight prefers deepseek-v4.1-flash writing with kimi-k3 reviewing (realised loss \$0.801); the rule picks deepseek-v4.1-flash with qwen3.8-max-0902 (realised \$0.829, 3.4% worse). The same rule priced with the fixed shape picks muse-spark-1.3 with qwen3.8-max-0902 (realised \$1.263, 57.5% worse). At $L = \$10$ the rule’s pick is unchanged and now costs 92.1% more than hindsight, while the fixed-shape variant is unchanged at 57.5% — the fixed shape is better here, by luck, and we say so. At $L \rightarrow \infty$ hindsight and the rule disagree again, hindsight taking the assignment with 0/11 failures and the rule the one with 0/4.

The reason the rule misses the hindsight winner is not noise in the bill; it is Assumption 1. Marginally over the ledger, kimi-k3 in the reviewing role fails on 6 of 31 runs (0.194) while qwen3.8-max-0902 fails on 7 of 59 (0.119), so an additive model prefers the latter at that role, correctly given its assumptions. The hindsight winner is a *pair*: kimi-k3 fails on 0 of 11 runs specifically alongside deepseek-v4.1-flash. An additive logit cannot represent that, and the pair term of Section 7 is exactly what would.

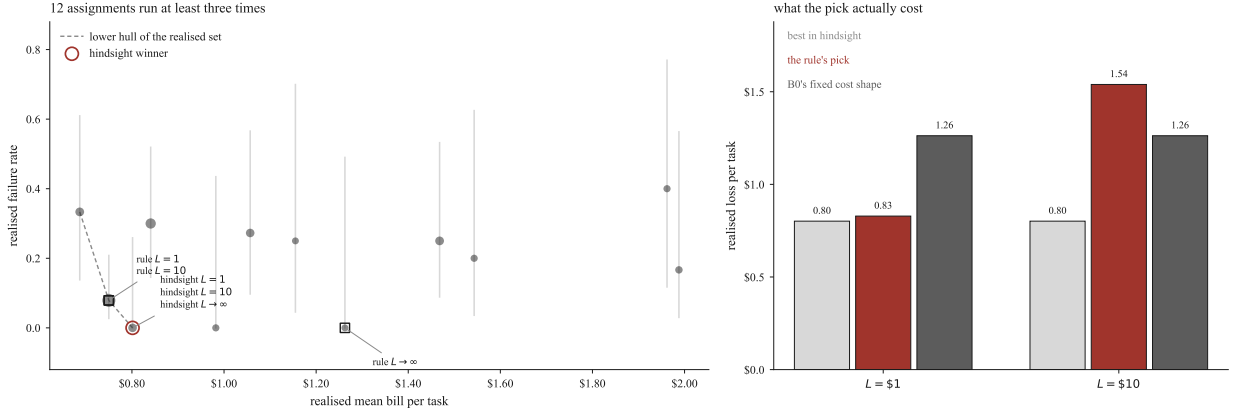


Figure 8: Measured. Left: the 12 assignments run at least three times, in realised mean bill against realised failure rate; marker area is the number of runs, vertical bars are Wilson intervals, and the dashed path is the lower hull of the realised set — the empirical version of Proposition 2. Circled is the assignment hindsight prefers at every stake; squared are the rule’s picks. Right: what those picks actually cost per task, against the best assignment in hindsight and against the same rule priced with B0’s fixed cost shape.

It should also be said plainly that the sample cannot resolve the difference. The hindsight winner’s failure rate is 0/11, Wilson interval $[0.000, 0.259]$; the rule’s pick is 3/38, interval $[0.027, 0.208]$. The intervals overlap over most of their length. By Proposition 7, separating two models at one role by $\Delta = 0.10$ at $\kappa = 30$ needs 114 graded runs *per model per role*; this ledger has 136 runs in total. At $L = \$1$ the bill term dominates and the rule is within a few cents of hindsight; at $L = \$10$ the failure term dominates and the ledger has nothing to say. That transition, rather than either number, is the finding.

13.4 E3: does the rule converge?

We replay the 136 graded runs in completion order. At step t the rule refits the quality model on the runs seen so far and picks among the assignments observed so far; it is charged the plug-in realised loss of the assignment it picks, and its *empirical regret* is the running excess over the best assignment in hindsight. Three rules are replayed: the proposed one; B1, which is the same quality model priced by the fixed cost shape; and B0, the scale-free knee on the fixed cost shape and the catalog score, which takes no stake and therefore draws the same picks at every L . Figure 9 shows the curves and Table 4 the slopes.

This is a replay, not a policy evaluation. The rule learns from the logged stream but does not control it, and it is scored against full-sample plug-in means; the quantity it measures is how fast the estimator converges to the hindsight-optimal assignment, not the regret of a policy that also chooses its own data. Section 14 says why the stronger claim is not available from these logs.

stake	rule	regret total	slope, by third of the replay		
			first	second	last
$L = \$1$	proposed	\$61.29	+0.896	+0.403	+0.027
	B1 (fixed shape)	\$65.64	+0.491	+0.461	+0.451
	B0 (knee)	\$31.43	+0.298	+0.181	+0.177
$L = \$10$	proposed	\$68.35	+0.448	+0.355	+0.580
	B1 (fixed shape)	\$73.62	+0.589	+0.461	+0.451
	B0 (knee)	\$44.03	+0.498	+0.181	+0.177
$L \rightarrow \infty$	all three	1.40	+0.022	+0.000	+0.000

Table 4: E3. Empirical regret against the best assignment in hindsight over 136 replayed runs, and its slope over each third. Dollars per task for the two finite stakes; excess failures per task for $L \rightarrow \infty$, where all three rules coincide after the sixth run and the hindsight optimum is zero failures.

At $L = \$1$ the rule converges and the baselines do not. Its per-task gap falls from \$0.90 to \$0.40 to \$0.027 across the three thirds — by the end it is within three cents per task of the assignment hindsight would have chosen, which is the 3.4% of E2 seen as a rate. B1’s gap is flat at \$0.45: with the same quality model and the same information, pricing by a fixed shape stops the rule converging at all. B0’s is flat at \$0.18 by construction, since after the front

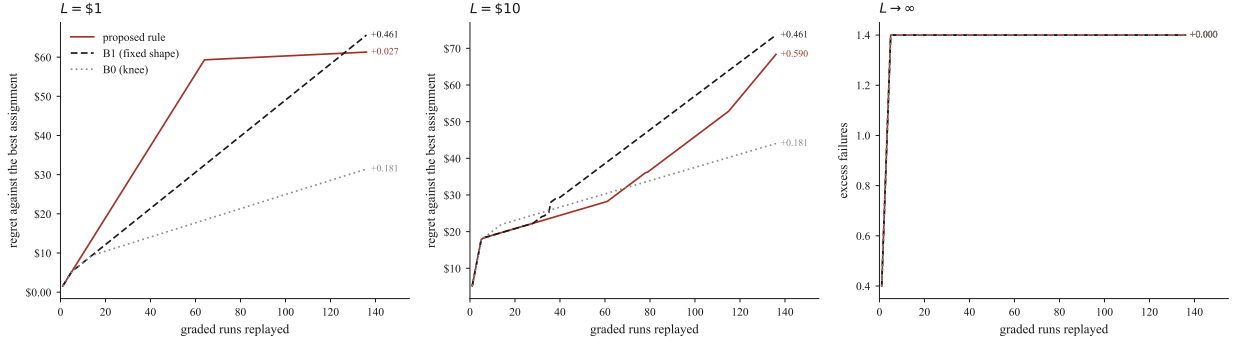


Figure 9: Measured. Empirical regret against the best assignment in hindsight, replaying 136 graded runs in completion order, at three stakes; annotations are the slope over the last third. At $L = \$1$ the proposed rule flattens to three cents per task while both baselines stay linear. At $L = \$10$ nothing flattens, for the reason Proposition 7 gives. At $L \rightarrow \infty$ the three rules coincide and the panel is uninformative; it is shown rather than omitted.

stabilises it holds one assignment forever. The proposed rule’s *cumulative* regret is still the larger of the three at the end of the replay, because of what it paid while learning; extrapolating the last third’s slopes, it overtakes B0 cumulatively after about 200 further tasks. We report that extrapolation as an extrapolation.

At $L = \$10$ it does not converge. The rule’s last-third slope is \$0.58 per task, higher than its second third. The cause is E2’s: at a stake of ten dollars the objective is dominated by a failure-rate difference of a few percentage points between two assignments whose Wilson intervals overlap, and 136 runs cannot order them. The rule is not oscillating — it settles on one assignment and holds it — it has settled on the wrong one and has no evidence that would move it. This is the regime Proposition 7 describes and it is the argument for pooling across sites, not for a different rule.

At $L \rightarrow \infty$ the experiment is uninformative. All three rules incur 1.4 excess failures over 136 runs against a hindsight optimum of zero, and all three curves are flat after the sixth run. With three assignments at exactly zero observed failures out of 11, 5 and 4 runs, there is nothing here to distinguish. We show the panel rather than omit it.

13.5 What the three experiments jointly support, and what they do not

They support the cost half of the thesis strongly and the quality half weakly, and the asymmetry is a function of how much data each half needs. A bill is observed exactly on every run, so 132 runs are enough to fit a per-role token shape and to show that doing so cuts prediction error by a factor of five and reorders the assignments. A grade is one bit per run, so 136 runs are nowhere near enough to order twelve assignments by failure rate, and at a stake where the failure term dominates the rule inherits that ignorance. The practical reading is that a deployment should expect the cost model to pay for itself immediately and the quality model to pay for itself only with pooled evidence — which is what Section 12 exists for, and what Proposition 7 predicted before any of this was measured.

14 Evaluation protocol

The bar. A rule is better if it picks assignments that are better on the cost–quality front, per role and per task, than the posterior-mean blend of B1, on the same data. Not: if it is more principled. Not: if it explores.

Off-policy estimation needs propensities that do not exist. For a logged history $\{(x_t, a_t, g_t)\}$ under a behaviour policy π_0 and a candidate π , the inverse-propensity estimator $\hat{V}_{\text{IPS}}(\pi) = \frac{1}{T} \sum_t \frac{\pi(a_t|x_t)}{\pi_0(a_t|x_t)} y_t$ and its doubly robust refinement [7] both require $\pi_0(a_t | x_t)$ to be recorded at the time of the pick. It is not recorded, and the reason is structural: the assignment pick is deterministic, so π_0 is a point mass and every importance weight is 0 or ∞ . *So the first requirement of this protocol is a logging change:* under the Thompson setting, record the sampled $\tilde{\theta}$ ’s seed and the resulting selection probability (estimable by replaying the draw R times at the stored posterior), and under the greedy setting record that the propensity is one, which makes the honest statement that no off-policy claim is possible from greedy logs.

Until then: replay and simulation, both labelled. Two admissible claims remain, and Section 13 makes exactly those. (a) *Replay*: recompute what each rule would have picked for each historical task and compare the picks’ bills exactly — bills are deterministic given the catalog and the shape, so this is an exact statement about cost with no counterfactual outcome in it — and score the quality term against plug-in means, labelled as such. (b) *Simulation*,

calibrated on the ledger, labelled as simulation in every caption, with its sensitivity to L , to the prior’s error and to the grade’s variance reported, as in Section 9. Neither is a randomised trial. A randomised trial over assignments is the right experiment and is the thing to build next; with $\Delta = 0.20$ and $\kappa = 30$ it needs about six graded runs per model per role (Proposition 7), which is within reach of a single team over a week.

15 Exposing the dial

The dial is one sentence: *what does it cost you when a task comes back wrong?* A hobby script, nothing much. A change a colleague will review, rather more. A change going near a release, whatever it takes.

preset	L	what it means
<i>frugal</i>	\$1	a failed task is an annoyance; spend nothing to avoid one
<i>balanced</i>	\$10	a failed task costs about ten minutes of a person
<i>max</i>	$\rightarrow \infty$	the best assignment on the front, whatever the bill
<i>custom</i>	any L	the number in dollars

Two of the presets are stakes in dollars; the third is the limit. By Proposition 3 the limit is well defined and cheap: as $L \rightarrow \infty$ the argmin of J is the last hull vertex a_K , the assignment with the lowest failure probability, with the bill as tie-break. That is what a superlative should mean. Naming a large finite number instead — fifty dollars, say — would be a fourth ordinary stake wearing a superlative’s label, and on any front whose top breakpoint L_K exceeds it, it would silently decline to buy the best assignment available.

Three properties follow from the machinery rather than from the copy.

Bills become predictable per band. Because the shortlist keeps at most k models per band (Section 6) and the bill is computed from a measured token shape (Section 8), the rule can state, before the run, the band it will run in and therefore the bill’s range within a factor of $\beta = 2$. A rule with a fixed shape can state a bill computed from a constant, which Section 13 shows is wrong by a median factor of 3.3 at the level of a run and 2.2 at the level of an assignment.

Every pick has a reason, and it is the posterior. “Why this assignment?” is answered by the two numbers in Eq. (5) and one from Eq. (6): this assignment costs $\$x$ and accepts p of the time; the next one up costs $\$y$ more and accepts p' ; at your stake of $\$L$ the upgrade is worth it above $\$L_j$ and you are below it. That sentence is a by-product of the posterior, not a separate explanation feature, and it cannot drift out of agreement with the pick because it is computed from the same numbers.

What exploration costs is a number the practitioner set. E is dollars per day. The system never spends more than E above what the greedy pick would have cost, and it reports how much of E it used.

16 Limitations

1. **One repository, one language, one window.** All 202 runs come from a single deployment, in one repository, inside a single 21-hour window. There are no external sites. Every measured number in this paper — the role shares, the proxy reliabilities, the cost spread, all of Section 13 — is a statement about that sample and about nothing else. The between-site variance σ_u^2 , on which the cap of Eq. (11) entirely depends, is unmeasured, because there is exactly one site.
2. **The outcome measured is a proxy, not the grade.** The historical runs predate an executable grader, so E2 and E3 use the reviewing role’s major-defect flag (with an empty artefact counted as a failure) in place of Definition 2. That proxy is endogenous in exactly the way Section 10 warns about — the reviewing role is part of the assignment being scored — and Section 4 measured that a related proxy is worth $\rho = 0.42$ of a grade. Every quality claim in Section 13 should be read through that factor. Repeating E2 and E3 against Definition 2 is the single highest-value next measurement.
3. **g is binary and only as good as the checks.** A change that builds, passes and is formatted, but does not do what the brief asked, scores 1. A repository with weak tests grades weakly. We regard this as the dominant threat to validity of the reward itself: it is honest and narrow, and the narrowness is not free. Extending g to a graded scale (fraction of acceptance criteria met) is straightforward statistically and hard to specify in a way that no model can influence.
4. **The objective is risk-neutral.** Eq. (5) optimises an expectation. A practitioner who cares about the worst task of the week wants a mean-variance or CVaR objective; $J_\eta(a) = \mathbb{E}[B] + L(1 - P) + \eta\sqrt{\text{Var}[B]}$ is a one-line change with a second dial, and a second dial is exactly what Section 5 argued against. We have not resolved this.

5. **The additive logit ignores interactions, and E2 shows it costing something.** The hindsight winner at every stake is a role pair the additive model cannot represent. The cascade captures the largest interaction we expect structurally; a pair term would capture more, at a cost in observations that Proposition 7 says this ledger cannot afford.
6. **E3 is a replay, not a policy evaluation.** The rule learns from a stream it did not choose and is scored against plug-in means. It measures estimator convergence, not policy regret, and the two coincide only if the logged design is close to uniform over the assignments — which it approximately is here, by construction of the logging experiment, but which we have not tested.
7. $k = 3, \beta = 2, \lambda = 4, h$ **are judgements.** They are marked as such in Table 2 and each should be replaced by an estimate. We report the pick’s sensitivity to λ and κ in the released results: the picks at $L = \$1$ and $L = \$10$ are unchanged over $\lambda \in \{0, 2, 4, 8\}$ and over $\kappa \in \{10, 30, 100\}$, while the pick at $L \rightarrow \infty$ moves at $\lambda = 0$ and again at $\kappa = 100$ — which is what one should expect of the stake at which the bill stops mattering and only the prior’s ordering of rarely-run assignments is left.
8. **Exploration was not shown to pay.** Section 9 reports a negative result. We have not shown that Thompson sampling inside J improves the rule in the regime it will mostly run in; we have shown where it is indispensable and where it is a tax, and set the default accordingly.
9. **No off-policy evidence is possible from these logs** (Section 14), and none is claimed.

17 Related work

The decision-theoretic core is bi-objective scalarisation and the classical fact that linear scalarisation reaches exactly the supported efficient set [11, 9]; Pareto bandits [6] address the case where no scalarisation is available, which is the case we deliberately leave by putting the exchange rate in dollars. The learner is a combinatorial bandit with semi-bandit feedback [2, 26, 25] under Thompson sampling [24, 18, 21]; information-directed sampling [19, 20] is the value-of-information alternative discussed in Section 9. The budget is the bandits-with-knapsacks constraint [1] without its algorithms. Non-stationarity is handled by variance inflation in the manner of Glicko-2 [12] and the local-level model [13], rather than by the discounting and sliding windows of [10]. Switching costs and their $T^{2/3}$ consequence are Dekel et al. [5]; anytime-valid testing at every boundary is Howard et al. [15]. Cascades over price tiers for language models are FrugalGPT [3]. Rater calibration is many-facet Rasch [17] on the continuous scale and Dawid–Skene [4] on the binary; shrinkage is Efron–Morris [8] and robustness is Huber [16]. Merging without a coordinator is the state-based CRDT [22]. Distributed and federated bandits [14, 23] describe the coordinated exploration we do not yet do. Off-policy evaluation is Dudík et al. [7].

18 Reproducibility

One extraction script turns the private run ledger and the public catalog snapshot into five text-free tables: one row per run, one per graded run, one per run with a per-role token and dollar split, and the catalog’s prices and indexes. It is the only script that reads a path outside the release directory. A second script draws every figure and computes every number quoted in Section 13 from those tables alone, and writes them to a plain-text results file that is released beside the figures; the sensitivity sweeps over λ and κ referenced in Section 16 are in it. Figures 1, 7, 8 and 9 are measured; Figures 2, 3, 4, 5 and 6 are simulation or closed form and say so in their captions. Nothing in the release carries task text, a file name, a brief or a reviewer’s sentence.

References

- [1] A. Badanidiyuru, R. Kleinberg, A. Slivkins. Bandits with knapsacks. *54th IEEE Symposium on Foundations of Computer Science (FOCS)*, pp. 207–216, 2013.
- [2] W. Chen, Y. Wang, Y. Yuan. Combinatorial multi-armed bandit: general framework and applications. *Proceedings of the 30th International Conference on Machine Learning (ICML)*, pp. 151–159, 2013.
- [3] L. Chen, M. Zaharia, J. Zou. FrugalGPT: how to use large language models while reducing cost and improving performance. arXiv:2305.05176, 2023.
- [4] A. P. Dawid, A. M. Skene. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society, Series C* 28(1):20–28, 1979.

- [5] O. Dekel, J. Ding, T. Koren, Y. Peres. Bandits with switching costs: $T^{2/3}$ regret. *Proceedings of the 46th ACM Symposium on Theory of Computing (STOC)*, pp. 459–467, 2014.
- [6] M. M. Drugan, A. Nowé. Designing multi-objective multi-armed bandits algorithms: a study. *International Joint Conference on Neural Networks (IJCNN)*, 2013.
- [7] M. Dudík, J. Langford, L. Li. Doubly robust policy evaluation and learning. *Proceedings of the 28th International Conference on Machine Learning (ICML)*, pp. 1097–1104, 2011.
- [8] B. Efron, C. Morris. Data analysis using Stein’s estimator and its generalizations. *Journal of the American Statistical Association* 70(350):311–319, 1975.
- [9] M. Ehrgott. *Multicriteria Optimization*, 2nd edition. Springer, 2005.
- [10] A. Garivier, E. Moulines. On upper-confidence bound policies for switching bandit problems. *Algorithmic Learning Theory (ALT)*, LNCS 6925, pp. 174–188, 2011.
- [11] A. M. Geoffrion. Proper efficiency and the theory of vector maximization. *Journal of Mathematical Analysis and Applications* 22(3):618–630, 1968.
- [12] M. E. Glickman. Dynamic paired comparison models with stochastic variances. *Journal of Applied Statistics* 28(6):673–689, 2001.
- [13] A. C. Harvey. *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge University Press, 1989.
- [14] E. Hillel, Z. Karnin, T. Koren, R. Lempel, O. Somekh. Distributed exploration in multi-armed bandits. *Advances in Neural Information Processing Systems (NeurIPS)* 26, 2013.
- [15] S. R. Howard, A. Ramdas, J. McAuliffe, J. Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics* 49(2):1055–1080, 2021.
- [16] P. J. Huber. Robust estimation of a location parameter. *Annals of Mathematical Statistics* 35(1):73–101, 1964.
- [17] J. M. Linacre. *Many-Facet Rasch Measurement*. MESA Press, Chicago, 1989.
- [18] D. Russo, B. Van Roy. Learning to optimize via posterior sampling. *Mathematics of Operations Research* 39(4):1221–1243, 2014.
- [19] D. Russo, B. Van Roy. Learning to optimize via information-directed sampling. *Advances in Neural Information Processing Systems (NeurIPS)* 27, 2014.
- [20] D. Russo, B. Van Roy. Learning to optimize via information-directed sampling. *Operations Research* 66(1):230–252, 2018.
- [21] D. Russo, B. Van Roy, A. Kazerouni, I. Osband, Z. Wen. A tutorial on Thompson sampling. *Foundations and Trends in Machine Learning* 11(1):1–96, 2018.
- [22] M. Shapiro, N. Preguiça, C. Baquero, M. Zawirski. Conflict-free replicated data types. *Stabilization, Safety, and Security of Distributed Systems (SSS)*, pp. 386–400, 2011.
- [23] C. Shi, C. Shen. Federated multi-armed bandits. *Proceedings of the AAAI Conference on Artificial Intelligence* 35(11):9603–9611, 2021.
- [24] W. R. Thompson. On the likelihood that one unknown probability exceeds another in view of the evidence of two samples. *Biometrika* 25(3/4):285–294, 1933.
- [25] S. Wang, W. Chen. Thompson sampling for combinatorial semi-bandits. *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pp. 5114–5122, 2018.
- [26] Z. Wen, B. Kveton, A. Ashkan. Efficient learning in large-scale combinatorial semi-bandits. *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, pp. 1113–1122, 2015.